

FIT 3182 2026 Assignment 2

Marking Guide

Marked By :				
Tasks	Sub Task	Criteria	Maximum Marks	Score (0 - 100%)
		Note: 1) Each criteria has requirements to obtain either a P, C, D, or HD. 2) To obtain a specific grade, you need to meet the requirements of the specific grade and the requirements of the grades(s) below it. 3) For instance, if you are aiming for HD in Task 1.1, you need to also meet the requirements for P, C and D in this task. 4) The percentage of the obtained grade will be multiplied with the maximum marks of a criteria to obtain the actual marks.		
Part 1. MongoDB Data Model (2 marks)	1.1	Collection Design a) Pass - Significant aspects are unclear. - Significant spelling and grammatical errors. b) Credit - Provide justifications with limited descriptions. - Few spelling errors present. c) Distinction - Good detail on justification with clear argument. - Quality of writing is at high standard. d) High Distinction - High-level of justification with clear argument and excellent description. - Excellent standard and quality of writing. No spelling and grammatical errors present.	1	
	1.2	Collection Relationship a) Pass - Significant aspects are unclear. - Significant spelling and grammatical errors. b) Credit - Provide justifications with limited descriptions. - Few spelling errors present. c) Distinction - Good detail on justification with clear argument. - Quality of writing is at high standard. d) High Distinction - High-level of justification with clear argument and excellent description. - Excellent standard and quality of writing. No spelling and grammatical errors present.	1	
Part 2. Streaming Application (16 marks)	2.1.1	Producer Implementation a) Pass - Implement Producers A and B. - Batching works every n seconds on matching batch IDs. - Minimal or no metadata provided. b) Credit - Implement Producers A, B and C. - Correct metadata (producer_id) is provided. c) Distinction and High Distinction - Metadata and timestamps are provided.	2	
	2.1.2	Streaming Join Logic a) Pass - Simple join on matching batch_id from A and B. - Drop records where batch_id mismatches. - Use a basic in-memory state to hold one batch per stream. b) Credit - Join on matching batch_id from producers A, B and C. c) Distinction - Join on records from producers A, B and C even when batch_id differ by a fixed offset. - Log dropped pairs. d) High Distinction - Semi-stream join across all 3 producers. e) High Distinction+ - Online join across all 3 producers.	6	
	2.1.3	Streaming Application - Sink Integration (MongoDB) a) Pass - Writes violation record to the MongoDB collection. b) Credit - Handle write failures with a simple retry. c) Distinction - Bulk writes for efficiency. d) High Distinction - Indexes for violation record.	2	
	2.1.4	Violation Detection a) Pass - Implement an instantaneous speed check against the camera limit. b) Credit - Add average speed check for the ending camera. - Both rules trigger violations correctly (instantaneous and average between cameras). c) Distinction and High Distinction - Parameterizable speed-limit thresholds per camera.	2	
	2.2.1	Data Visualisation - Correctness of Data Processing, Plot Design and Clarity a) Pass - Plots runs but with some errors. - Two separate line charts produced. - Labelled axes. b) Credit - Plots run without error. - Title, axis labels, and legends present. - Different line styles or markers for clarity. c) Distinction - Dual-axis or subplots arranged side-by-side comparison. - Annotates on interesting points. d) High Distinction - Consistent styling and high quality visualisation.	2	
	2.2.2	Data Visualisation - Interesting Point Annotation a) Pass - Marks one extreme point. b) Credit - Create interesting points based on the specification. c) Distinction - Highlights additional insights (sudden spikes or drops). - Uses callouts or shaded regions for periods of interest. d) High Distinction - Dynamically computes and labels percentiles.	2	

Part 3: Documentation (2 marks)	3.1	Code Comments a) Pass - Sparse comments. b) Credit - Comments explain most non-obvious steps. - Functions have brief docstrings describing input and output. c) Distinction - Comments clearly explain algorithmic intent. - All functions/methods have full docstrings. d) High Distinction - Docstrings follow standard format.	1	
	3.2	Narrative Explanation (using markdown) a) Pass - Trivial screenshots of code and minimal explanation in markdown. b) Credit - Flowchart or sequence diagram to describe the implemented algorithms. c) Distinction - A sufficient explanation of the algorithms or methods used in the assignment, in the form of written or illustrated. - References to external concepts or docs, where helpful. d) High Distinction - Detailed or comprehensive explanation of the end-to-end algorithm, both in written, illustrated or tabulated form. For instance, a table summarising key parameters (e.g., window size for join).	1	
Final Grade				
Contribution to Final Marks				